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Module 3: Exchange and Transport

OCR A A-Level Biology (H420)

3.1 Exchange Surfaces

Surface Area to Volume Ratio

As organisms increase in size, their volume increases more rapidly than their surface area (volume increases as the cube of linear dimensions, while surface area increases as the square). This means that large, multicellular organisms cannot rely solely on diffusion across their body surface to exchange substances with the environment. They require specialised exchange surfaces and transport systems to maintain adequate rates of exchange.

> **Fick's Law of Diffusion:** Rate of diffusion is proportional to (surface area x concentration difference) / thickness of exchange surface. This relationship explains why effective exchange surfaces are adapted to maximise surface area, maintain steep concentration gradients, and minimise diffusion distance.

Features of Effective Exchange Surfaces

All exchange surfaces share common adaptations that maximise the rate of diffusion according to Fick's Law:

- **Large surface area** — achieved through folding, branching, or flattening (e.g., alveoli, villi, root hair cells)
- **Thin barrier** — minimises diffusion distance (e.g., alveolar epithelium is one cell thick, approximately 0.5 micrometres)
- **Steep concentration gradient maintained** — through ventilation (breathing), blood flow, or active removal of diffused substances

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- **Good blood supply** — carries substances to and from the exchange surface, maintaining concentration gradients

Gas Exchange in Mammals (Lungs)

The mammalian gas exchange system consists of the trachea, bronchi, bronchioles, and alveoli. The trachea and bronchi are supported by incomplete C-shaped rings of cartilage that prevent collapse during pressure changes while allowing the oesophagus to expand during swallowing. The airways are lined with ciliated epithelium and goblet cells that produce mucus, trapping particles and sweeping them towards the throat (mucociliary escalator).

Alveoli are the sites of gas exchange. Oxygen diffuses from the alveolar air space into the blood, while carbon dioxide diffuses in the opposite direction. The adaptations of alveoli for efficient gas exchange include:

- Enormous total surface area (approximately 70 square metres in human lungs due to 300 million alveoli)
- Extremely thin walls (single layer of squamous epithelial cells, approximately 0.5 micrometres)
- Dense capillary network surrounding each alveolus with walls only one cell thick
- Ventilation maintains a steep concentration gradient for oxygen and carbon dioxide
- Continuous blood flow through pulmonary capillaries carries oxygen away and brings carbon dioxide
- Moist lining (surfactant) allows gases to dissolve before diffusing across the membrane

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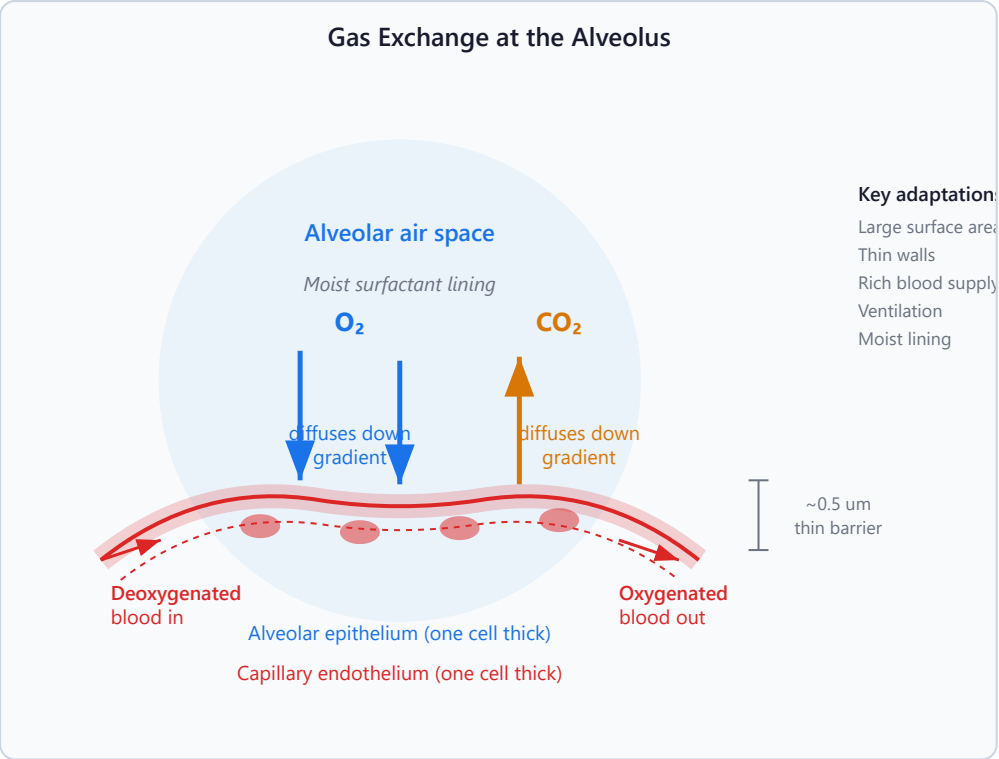

Ventilation is achieved by the action of the diaphragm and intercostal muscles. During inspiration, the diaphragm contracts and flattens while external intercostal muscles contract to raise the ribcage outward and upward, increasing thoracic volume, decreasing intrapulmonary pressure below atmospheric pressure, and causing air to flow in. Expiration at rest is largely passive: muscles relax, elastic recoil of lung tissue decreases thoracic volume, and air is pushed out.

Gas Exchange in Fish

Fish exchange gases using gills, which consist of gill filaments bearing numerous lamellae that provide a large surface area. Water flows over the lamellae in the opposite direction to blood flow within them, establishing a **countercurrent exchange system**. This maintains a concentration gradient for oxygen along the entire length of the lamella, achieving up to 80% oxygen extraction from water (compared to approximately 25% in concurrent flow).

The countercurrent mechanism works because at every point along the lamella, the water adjacent to the blood has a higher oxygen concentration than the blood. This means a diffusion gradient exists along the entire length of the exchange surface, not just at the beginning. Without this arrangement, equilibrium would be reached partway along and net diffusion would cease.

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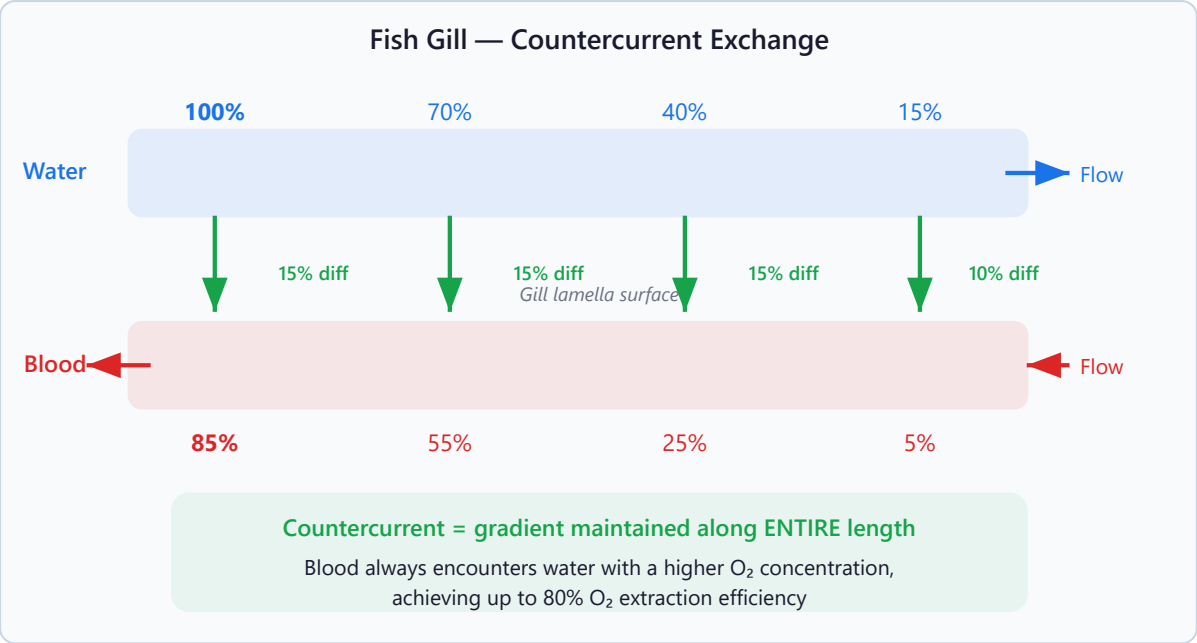

Gas Exchange in Insects

Insects have a tracheal system consisting of spiracles (openings on the body surface that can open and close to regulate water loss), tracheae (tubes reinforced with chitin rings to prevent collapse), and tracheoles (tiny blind-ended tubes that penetrate individual cells). Oxygen diffuses directly from the air in tracheoles to respiring cells, and carbon dioxide diffuses in the reverse direction. This system is effective for small organisms but limits maximum body size because diffusion is only efficient over short distances.

During high metabolic demand (e.g., flight), muscles contract to ventilate the tracheal system, and tracheole fluid is withdrawn by osmosis (as lactic acid accumulates in active cells, lowering water potential), bringing air closer to respiring tissues and reducing diffusion distance.

Gas Exchange in Plants

Leaves exchange gases primarily through stomata, which are pores in the epidermis (mainly on the lower surface) controlled by paired guard cells. When guard cells are turgid (due to potassium ion uptake and subsequent osmotic water entry), they bow apart, opening the stomatal pore. When guard cells lose turgor, the stomata close, reducing water loss by transpiration.

Inside the leaf, the spongy mesophyll layer has a large internal surface area with air spaces between cells, allowing efficient gas exchange. Carbon dioxide dissolves in the moist cell walls before diffusing into cells for photosynthesis, while oxygen produced by photosynthesis diffuses out through the same route.

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Exam Tip: When explaining gas exchange adaptations, always link structural features to Fick's Law components. For example: "The alveolar wall is one cell thick (thin barrier), which minimises diffusion distance, increasing the rate of gas exchange according to Fick's Law."

3.2 Transport in Animals

The Circulatory System

Mammals have a closed, double circulatory system. Blood flows through blood vessels (closed) and passes through the heart twice per complete circuit (double). The pulmonary circulation carries deoxygenated blood from the right ventricle to the lungs for gas exchange and returns oxygenated blood to the left atrium. The systemic circulation pumps oxygenated blood from the left ventricle to the body tissues and returns deoxygenated blood to the right atrium.

The advantage of a double circulation is that blood pressure is restored by the heart between the pulmonary and systemic circuits. The left ventricle can generate high pressure for efficient delivery to body tissues without subjecting the delicate pulmonary capillaries to this pressure. This allows mammals to maintain the high metabolic rates needed for endothermy.

The Heart

The heart is a muscular pump divided into four chambers: two atria (thin-walled, receive blood) and two ventricles (thick-walled, pump blood out). The left ventricular wall is thicker than the right because it must generate sufficient pressure to pump blood around the entire systemic circulation, whereas the right ventricle only pumps blood the short distance to the lungs.

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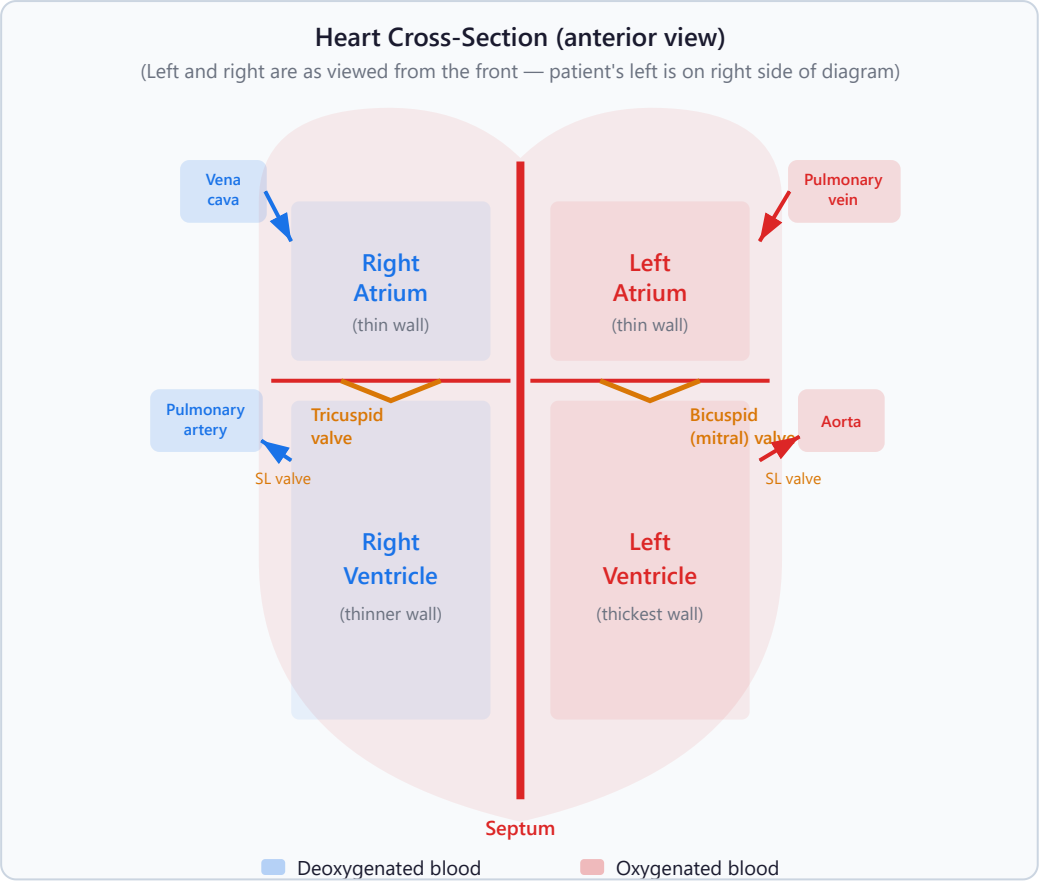

The cardiac cycle consists of atrial systole (atria contract, forcing blood into ventricles through open atrioventricular valves), ventricular systole (ventricles contract, AV valves close to prevent backflow, semilunar valves open as ventricular pressure exceeds arterial pressure, blood is ejected into arteries), and diastole (all chambers relax, heart fills passively with blood, semilunar valves close as arterial pressure exceeds ventricular pressure).

Heart rate is controlled by the sinoatrial node (SAN), the natural pacemaker located in the right atrial wall. The SAN generates electrical impulses that spread across the atria causing atrial contraction. The impulse is delayed at the atrioventricular node (AVN) to allow the atria to empty before the ventricles contract. From the AVN, impulses travel rapidly down the bundle of His and Purkyne fibres to the ventricular apex, causing contraction from the bottom upward to efficiently eject blood into the arteries.

> **Cardiac Output:** The volume of blood pumped by the heart per unit time. Cardiac output = stroke volume x heart rate. Stroke volume is the volume of blood ejected per beat (typically 70 cm3 at rest). Heart rate is beats per minute (typically 70 bpm at rest). Therefore, resting cardiac output is approximately 4900 cm3 per minute (4.9 dm3/min).

Blood Vessels

Vessel	Structure	Function
Arteries	Thick walls with elastic tissue and smooth muscle; narrow lumen relative to diameter; no valves (except at base of aorta and pulmonary artery)	Carry blood away from heart at high pressure; elastic recoil smooths pulsatile flow; smooth muscle allows vasoconstriction/vasodilation
Arterioles	Proportionally more smooth muscle than arteries; narrow lumen	Control blood distribution to tissues; vasoconstriction increases resistance, redirecting blood to active tissues
Capillaries	One cell thick (endothelium only); very narrow lumen (approximately 7-10 micrometres); no smooth muscle or elastic tissue; pores between cells	Site of exchange between blood and tissues; thin walls minimise diffusion distance; narrow lumen forces red blood cells through in single file, maximising exposure to surrounding tissue
Veins	Thin walls with less smooth muscle and elastic tissue; wide lumen; contain valves	Return blood to heart at low pressure; valves prevent backflow; wide lumen reduces resistance to flow; skeletal muscle pump and respiratory pump aid venous return

Haemoglobin and Oxygen Transport

Haemoglobin is a quaternary protein consisting of four polypeptide subunits (two alpha and two beta chains), each containing a haem prosthetic group with an iron (Fe^{2+}) ion at its centre. Each haem group can reversibly bind one oxygen molecule, so one haemoglobin molecule can carry a maximum of four oxygen molecules (eight oxygen atoms).

The **oxygen dissociation curve** is a sigmoid (S-shaped) curve that shows the relationship between the partial pressure of oxygen (pO_2) and the percentage saturation of haemoglobin. The sigmoid shape results from cooperative binding: when the first oxygen molecule binds to haemoglobin, it

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causes a conformational change that increases the affinity of the remaining haem groups for oxygen. Similarly, as oxygen is released, the remaining oxygen molecules are released more readily.

The **Bohr effect** describes the shift of the oxygen dissociation curve to the right in the presence of higher carbon dioxide concentration (lower pH). In active tissues where respiration rates are high, carbon dioxide accumulates, lowering pH. This decreases haemoglobin's affinity for oxygen, promoting oxygen release where it is most needed. In the lungs, where carbon dioxide concentration is low, haemoglobin has high affinity for oxygen, promoting oxygen loading.

Fetal haemoglobin (HbF) has a higher oxygen affinity than adult haemoglobin (HbA) — its dissociation curve is to the left of adult haemoglobin. This is because fetal haemoglobin has a different structure (two alpha and two gamma subunits instead of two alpha and two beta) that binds oxygen more tightly. This allows the fetus to extract oxygen from maternal haemoglobin across the placenta, where the partial pressure of oxygen is relatively low. At the same pO_2, fetal haemoglobin is more saturated than maternal haemoglobin, ensuring oxygen transfer from mother to fetus.

Myoglobin is a single-subunit oxygen-binding protein found in muscle tissue. It has an even higher oxygen affinity than haemoglobin (curve further to the left), acting as an oxygen store in muscles that releases oxygen only when oxygen levels drop very low during intense exercise.

Carbon dioxide transport: CO_2 is transported in the blood in three ways: dissolved in plasma (~5%), bound to haemoglobin as carbaminohaemoglobin (~10%), and as hydrogencarbonate ions (~85%). In red blood cells, the enzyme **carbonic anhydrase** catalyses: $CO_2 + H_2O \leftrightarrow H_2CO_3 \leftrightarrow H^+ + HCO_3^-$. The HCO_3^- ions diffuse out of the red blood cell into the plasma via a chloride shift (Cl^- ions move into the cell to maintain electrical neutrality). This process is reversed in the lungs, where CO_2 is released for exhalation.

> **Exam Tip:** When interpreting dissociation curves, remember that a shift to the right means lower affinity (oxygen released more readily), and a shift to the left means higher affinity (oxygen binds more readily and is released less easily). Always relate shifts to biological context.

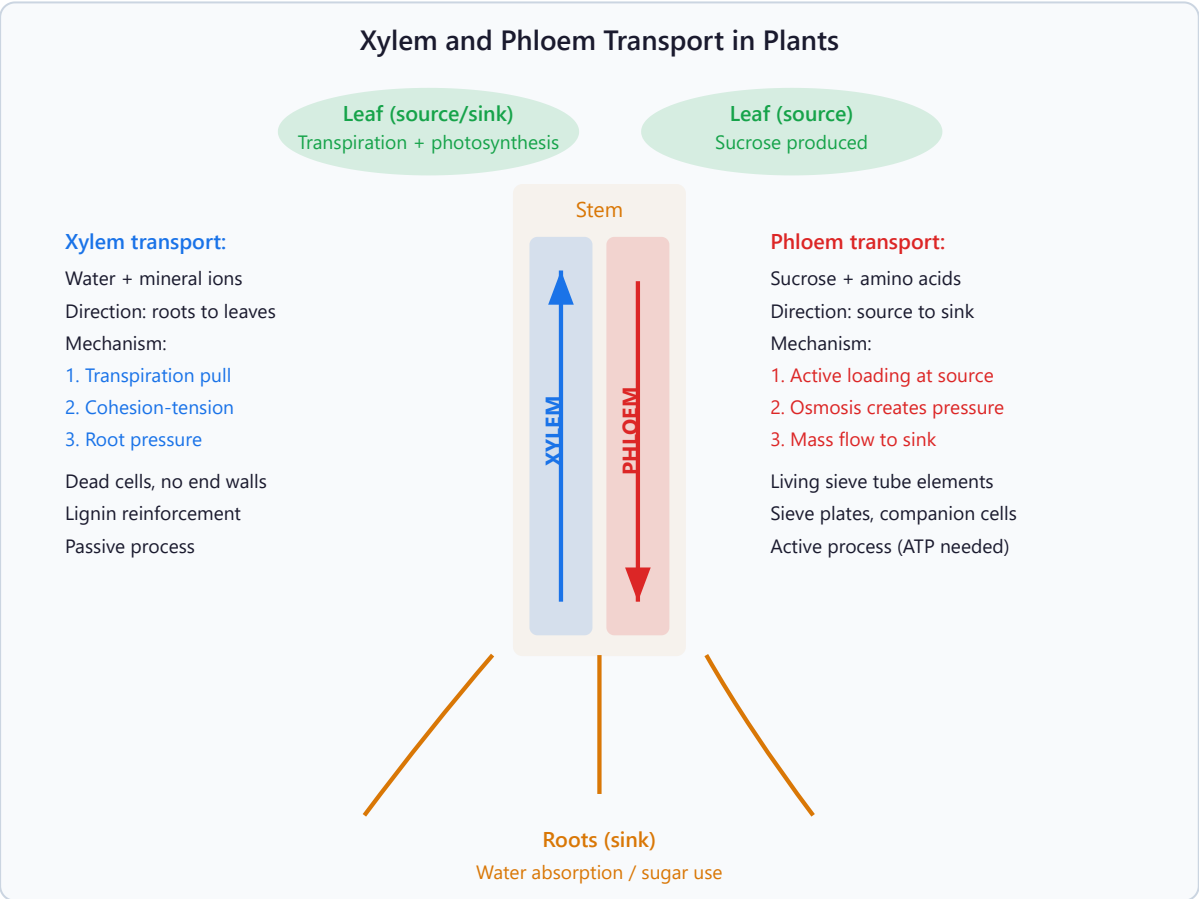
3.3 Transport in Plants

Xylem and Water Transport

Xylem vessels transport water and dissolved mineral ions from roots to leaves. They are formed from dead cells arranged end-to-end with no end walls, creating continuous hollow tubes. The walls are reinforced with lignin, which provides structural support, prevents collapse under negative pressure, and waterproofs the vessel walls.

Water moves through the plant via the **transpiration stream**, driven by three mechanisms:

- **Root pressure** — active transport of mineral ions into root xylem lowers water potential, drawing water in by osmosis, creating a small positive pressure
- **Cohesion-tension theory** — water evaporates from mesophyll cell surfaces into leaf air spaces (transpiration), creating tension (negative pressure) that pulls a continuous column of water up through the xylem. Cohesion between water molecules (due to hydrogen bonds) maintains the unbroken water column. Adhesion between water molecules and xylem vessel walls provides additional support.
- **Capillary action** — adhesion between water and the narrow xylem walls contributes to water movement, particularly in small-diameter vessels



Transpiration

Transpiration is the loss of water vapour from plant surfaces, primarily through stomata in leaves. It is a consequence of gas exchange (stomata must be open for carbon dioxide entry) rather than a beneficial process in itself, though it does contribute to cooling and mineral transport.

Factors affecting transpiration rate:

- **Light intensity** — stomata open in light for photosynthesis, increasing transpiration
- **Temperature** — higher temperature increases kinetic energy of water molecules, increasing evaporation rate and the water potential gradient between leaf and atmosphere
- **Humidity** — low humidity increases the water potential gradient between the moist leaf interior and dry external air, increasing transpiration rate
- **Wind speed** — wind removes water vapour from around the leaf surface, maintaining a steep diffusion gradient

Xerophytic plants (adapted to dry conditions) reduce water loss through thick waxy cuticles, sunken stomata (creating a humid microclimate that reduces the diffusion gradient), rolled leaves (trapping moist air), reduced leaf surface area (needles or spines), and extensive root systems to maximise water absorption.

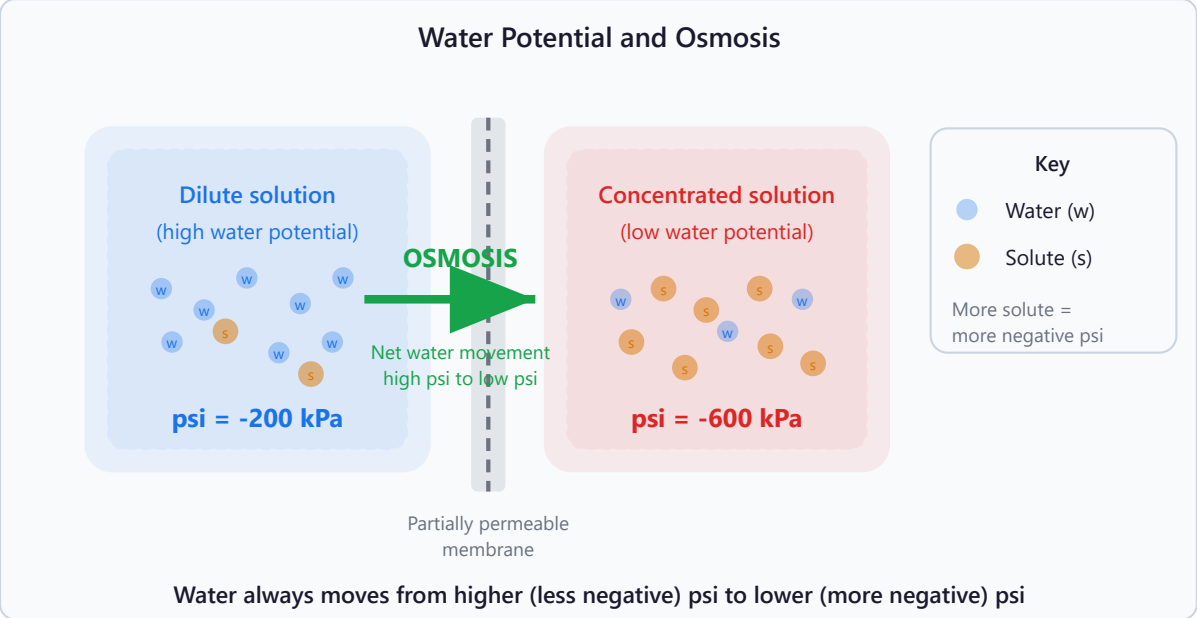
Phloem and Translocation

Phloem transports organic solutes (mainly sucrose and amino acids) from sources (where sugars are produced or mobilised, such as photosynthesising leaves or storage organs releasing reserves) to sinks (where sugars are used or stored, such as growing roots, developing fruits, or storage organs accumulating reserves).

Phloem tissue consists of sieve tube elements and companion cells. Sieve tube elements are living cells but lack nuclei, ribosomes, and most organelles to maximise space for flow. They are connected end-to-end through sieve plates (perforated end walls). Companion cells are metabolically active cells connected to sieve tube elements by numerous plasmodesmata, providing the ATP and proteins needed for sieve tube function.

> **Mass Flow Hypothesis (Pressure Flow):** The widely accepted explanation for translocation. At the source, sucrose is actively loaded into phloem sieve tubes (via companion cells using ATP-powered H+ cotransport), lowering the water potential inside the sieve tube. Water enters by

osmosis from adjacent xylem, creating high hydrostatic pressure. At the sink, sucrose is actively unloaded (or used in respiration), raising water potential in the sieve tube. Water leaves by osmosis. The pressure difference between source (high pressure) and sink (low pressure) drives mass flow of sap from source to sink.



Evidence supporting the mass flow hypothesis includes: sieve tube sap is under positive pressure (as shown by aphid stylet experiments, where sap exudes from severed stylets); flow rates measured are consistent with the hypothesis; radioactive tracer experiments show sugars move from source to sink; and ringing experiments (removing phloem but not xylem) cause sugars to accumulate above the ring.

Evidence that challenges the model includes: flow occurs in both directions simultaneously in adjacent sieve tubes; the mechanism for loading and unloading at sources and sinks requires active processes not fully explained by simple mass flow; and sieve plates would appear to impede flow (though they may function to prevent damage spreading).

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